Amrita School of AI
COIMBATORE CAMPUS

WEEK 11 · LAB

Lab 7: Domain Decomposition and Halo Exchange

Split a grid across ranks, exchange halos, and confirm the surface-to-volume prediction with your own measurements.

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DURATION
One 2-hour block

PREREQUISITE
Lab 6, Exercise 11

SUBMIT
lab07-<ROLLNO>.pdf

MARKS
12

OFFERING
Odd Semester 2026-27

The point of this lab

Exercise 11 had you write a 1-D halo exchange. This lab takes it to two dimensions, which is where the interesting decision appears: **how** you cut the domain changes how much you communicate, and the difference grows with the machine.

Part 1: 1-D scaling (3 marks)

Start from your Exercise 11 solution.

1.1 Run a strong-scaling study at n = 4 000 000, 2000 steps, on P = 1, 2, 4, 8, 16, 32 ranks. Plot speedup and efficiency.

1.2 Efficiency will fall. Compute the Karp-Flatt e at each point. Is it constant or rising? What does that say about the cause?

1.3 Now instrument the code: time the communication and the computation separately. Plot the **communication fraction** against P. Does it match your answer to 1.2?

> **DEFINITION · WHY THE RATIO WORSENS**
>
> Each rank's computation is proportional to its share of the domain, which falls as $1/P$. Its communication is two halo values, which does **not** fall. So the ratio of communication to computation grows with P, and eventually dominates.

Part 2: 2-D decomposition (5 marks)

Extend to a 2-D grid with a 5-point stencil, decomposed as a 2-D grid of ranks.

2.1 Use `MPI_Dims_create` and `MPI_Cart_create` to build a Cartesian communicator, and `MPI_Cart_shift` to find your four neighbours. Explain in two sentences what these buy you over computing neighbour ranks by hand.

2.2 Exchange all four halos with non-blocking calls. The north and south halos are contiguous rows; the east and west halos are **columns**, which are not contiguous. Handle them either with a manual pack/unpack buffer or with `MPI_Type_vector`. Say which you chose and why.

2.3 Verify correctness against a serial reference at several rank counts, including a count that does not factor evenly, such as 6 and 7.

2.4 Corner values: does your stencil need them? Explain what happens at the corners of each subdomain and whether a diagonal exchange is required for a 5-point stencil.

Part 3: The prediction (4 marks)

For an $N \times N$ grid on P ranks:

$$\text{1-D strips: ratio} = 2P / N \qquad \text{2-D blocks: ratio} = 4\sqrt{P} / N$$

3.1 For $N = 2048$ and $P = 4, 16, 64$, compute the predicted communication to computation ratio for both decompositions. Tabulate.

3.2 Measure the actual communication fraction for both, at the same points.

3.3 Plot predicted against measured. Do they agree in **trend**? In absolute value? If not, name two effects the simple model omits.

3.4 At what P does 2-D become worth its extra complexity? Answer with your own numbers.

BEYOND THE SYLLABUS · WHY THIS ARGUMENT DECIDES REAL CODE

The 2-D advantage grows as $\sqrt{P}$. At 64 ranks it is about four times less communication for identical computation; at 4096 ranks it is thirty-two times. This is why production solvers decompose in every dimension available, and why 3-D decomposition exists for 3-D problems.

What to submit

`lab07-<ROLLNO>.pdf`: the 1-D scaling study with the communication-fraction plot, your 2-D implementation as source, the correctness evidence at awkward rank counts, the prediction-versus-measurement table and plot, and your answer to 3.4.

MARKS	FOR
3	Part 1, including the communication-fraction instrumentation
5	Part 2: a correct 2-D exchange, verified at non-factoring rank counts
4	Part 3: prediction computed, measured, and the discrepancy explained